EXERCISES List 1 October 6, 2026

Simulations and algorithmic applications of Markov chains

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Probability

1. Let T be a random variable taking non-negative integer values. Show that

$$\mathbb{E}T = \sum_{k=0}^{\infty} \mathbb{P}(T > k).$$

2. (Markov's inequality). Let X be a non-negative random variable. Prove that, for every $r > 0$,

$$\mathbb{P}(X \geq r) \leq \frac{\mathbb{E}X}{r}.$$

3. (Chebyshev's inequality). Let X be a random variable with finite mean μ and variance $\sigma^2 < \infty$. Prove that, for every $r > 0$,

$$\mathbb{P}(|X - \mu| \geq r) \leq \frac{\sigma^2}{r^2}.$$

4. Show that

(a) $1 + x \leq e^x$ for every $x \in \mathbb{R}$;

(b) $\ln(1+x) \geq \dfrac{x}{1 + x/2}$ for every $x \geq 0$.

Probability: concentration inequalities

5. (Chernoff's inequalities). Let $X_1, \ldots, X_n$ be independent Bernoulli random variables, with $\mathbb{P}(X_i = 1) = p_i = 1 - \mathbb{P}(X_i = 0)$. Let

$$S = \sum_{i=1}^{n} X_i, \qquad \mu = \mathbb{E}S = \sum_{i=1}^{n} \mathbb{E}X_i = \sum_{i=1}^{n} p_i.$$

Prove that, for every $\varepsilon > 0$,

$$\mathbb{P}(S > (1+\varepsilon)\mu) \leq \left(\frac{e^{\varepsilon}}{(1+\varepsilon)^{1+\varepsilon}} \right)^{\mu},$$

and, for every $\varepsilon \in (0,1)$,

$$\mathbb{P}(S < (1-\varepsilon)\mu) \leq \left(\frac{e^{-\varepsilon}}{(1-\varepsilon)^{1-\varepsilon}}\right)^{\mu}.$$

6. Show that Exercise 5 implies that, for $\varepsilon \in (0,1)$,

$$\mathbb{P}(S < (1-\varepsilon)\mu) \leq e^{-\varepsilon^2\mu/2},$$
$$\mathbb{P}(S > (1+\varepsilon)\mu) \leq e^{-\varepsilon^2\mu/(2+\varepsilon)}.$$

7. Using the previous exercises, prove the following statement. Let $Y_1, \ldots, Y_R$ be i.i.d. indicator random variables with $p = \mathbb{E}Y_i > 0$, and let

$$\widehat{Y}_R = \frac{1}{R}\sum_{i=1}^{R} Y_i.$$

Fix $\varepsilon, \delta \in (0,1)$. If

$$R \geq \frac{3\ln(2/\delta)}{\varepsilon^2 p},$$

then

$$\mathbb{P}\big(|\widehat{Y}_R - p| \geq \varepsilon p\big) \leq \delta.$$

Markov chains

Distributions are row vectors. We write $\boldsymbol{\mu}^{(k)} = \boldsymbol{\mu}\mathbf{P}^k$ for the distribution after k steps, starting from $\boldsymbol{\mu}$.

8. Consider a simple random walk on $\mathbb{E} = \{1,2,3,4\}$ with transition matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \end{pmatrix}$$

and initial distribution $\boldsymbol{\mu} = (1,0,0,0)$.

(a) Compute $\boldsymbol{\mu}^{(3)}$.

(b) Compute $\boldsymbol{\mu}^{(n)}$ by guessing a formula and proving it by induction.

9. Consider a chain on $\mathbb{E} = \{\mathbf{e}_1, \mathbf{e}_2\} = \{\text{rainy}, \text{sunny}\}$. After a rainy day, the next day is rainy with probability $3/4$; after a sunny day, the next day is sunny with probability $3/4$. Thus

$$\mathbf{P} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

(a) Compute $\boldsymbol{\mu}^{(2)}$ for $\boldsymbol{\mu} = (0,1)$.

(b) Compute $\boldsymbol{\mu}^{(n)}$ for $\boldsymbol{\mu} = (0,1)$ using diagonalization of $\mathbf{P}$.

(c) Using part (b), compute $\lim_{n\to\infty} \boldsymbol{\mu}^{(n)}$ for any initial distribution.

10. As in Exercise 9, use the state order rainy, sunny, but now let

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 \\ 1/10 & 9/10 \end{pmatrix}.$$

Give a formula for $\boldsymbol{\mu}^{(k)}$ for each initial distribution:

$$\text{(a) } \boldsymbol{\mu} = (1,0), \qquad \text{(b) } \boldsymbol{\mu} = (1/2,1/2), \qquad \text{(c) } \boldsymbol{\mu} = (1/6,5/6).$$

Is there anything special about one of these cases? What does $\boldsymbol{\mu}^{(k)}$ converge to in each case as $k \to \infty$?

Markov chains: transitions and distances

11. Let $\mathbf{X} = (X_0, X_1, \ldots)$ be a time-homogeneous Markov chain with transition matrix $\mathbf{P}$ and state space $\mathbb{E} = \{\mathbf{e}_1, \ldots, \mathbf{e}_M\}$, so that

$$\mathbb{P}(X_{n+1} = \mathbf{e}_j \mid X_n = \mathbf{e}_i) = \mathbf{P}(\mathbf{e}_i, \mathbf{e}_j).$$

Show that

$$\mathbb{P}(X_{n+m} = \mathbf{e}_j \mid X_n = \mathbf{e}_i) = \mathbf{P}^m(\mathbf{e}_i, \mathbf{e}_j).$$

Conditional probabilities are considered whenever the conditioning event has positive probability.

12. For top-to-random shuffling with $n = 3$ cards, give the transition matrix $\mathbf{P}$, draw the transition graph, and compute $\mathbf{P}^2$. One step takes the top card and inserts it into one of the n positions, chosen uniformly at random.

13. For probability distributions $\boldsymbol{\mu}, \boldsymbol{\nu}$ on $\mathbb{E} = \{\mathbf{e}_1, \ldots, \mathbf{e}_M\}$, define

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) = \frac{1}{2} \sum_{\mathbf{e} \in \mathbb{E}} |\mu(\mathbf{e}) - \nu(\mathbf{e})|.$$

Compute this distance between the distributions $\mathrm{Bin}(4, 1/2)$ and $\mathrm{Bin}(4, 1/4)$.

14. On $\mathbb{E} = \{1, 2\}$, let

$$\boldsymbol{\mu}^{(k)} = \left(\frac{1}{6} + \frac{1}{3} \left(\frac{2}{5} \right)^k, \frac{5}{6} - \frac{1}{3} \left(\frac{2}{5} \right)^k \right),$$
$$\boldsymbol{\nu} = \left(\frac{1}{6}, \frac{5}{6} \right).$$

(a) Compute $d_{TV}(\boldsymbol{\mu}^{(k)}, \boldsymbol{\nu})$.

(b) For which integers $k \geq 0$ is this distance smaller than 0.01?

15. Consider n cards. Let $\boldsymbol{\mu}$ be the uniform distribution on all $n!$ permutations, and let $\boldsymbol{\nu}$ be the uniform distribution on all permutations with a specified card in the first position. Compute $d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu})$.